



# Synchronized stationary distribution of hybrid stochastic coupled systems with applications to coupled oscillators and a Chua's circuits network

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## Abstract

In this paper, the existence of synchronized stationary distribution for hybrid stochastic coupled systems (HSCSs) (here, also known as stochastic coupled systems with Markovian switching) is concerned. By the existence theory of stationary distribution as well as Lyapunov method and graph theory, two kinds of sufficient criteria are presented to promise the existence of synchronized stationary distribution for HSCSs. Our results exhibit that the existence region of synchronized stationary distribution has a close relationship with the intensity of stochastic perturbation. And when stochastic perturbation vanishes, synchronized stationary distribution will become complete synchronization. Then the proposed theoretical results are successfully applied to stochastic coupled oscillators and a Chua's circuits network. Some existence criteria of synchronized stationary distribution are also obtained. The corresponding numerical simulations are carried out to verify the validity of the theoretical results.

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## 1. Introduction

As a special class of stochastic hybrid systems, stochastic systems with Markovian switching are dynamical systems driven by a continuous-time Markov process [1]. This kind of stochastic hybrid systems has been widely used to model many practical systems where

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abrupt changes may appear in the structure and parameters caused by phenomena such as component failures or repairs. And many interesting results have been reported, see [1–4] for example. On the other hand, many complex systems in real life are coupled by some subsystems, such as neural networks [5–8], multi-group models [9], multi-agent systems [10] and so on. Hence, it is meaningful to study the dynamical properties of hybrid stochastic coupled systems (HSCSs), such as stability [11], synchronization [12] and so on.

Up to now, there have been a multitude of research results on HSCSs pertaining to the issue of synchronization, see [13–17] and the references therein. It should be mentioned here that stochastic perturbation intensity in above literature is related to the state of error system, i.e., when error tends to zero, perturbation intensity tends to zero as well. However, this situation does not always happen. For example, some environmental factors can be seen as constant stochastic perturbation. Constant stochastic perturbation will destroy the equilibrium of error system and will make that the studied system cannot achieve complete synchronization. In this case, what interesting dynamical phenomena will appear? In fact, for the study of stability theory of equilibrium for error system, when equilibrium is destroyed by stochastic perturbation, it may cause a weak stability phenomenon, which is named as stationary distribution in many literature [18–28]. In the study of synchronization, we call the phenomenon as synchronized stationary distribution, if error system exists a stationary distribution. Compared with the general meaning of synchronization in most of literature, synchronized stationary distribution is a weaker form of synchronization. Actually, synchronized stationary state was firstly proposed in [29] to study a class of Kuramoto model with specific coupling form in physics. However, due to the complexity of HSCSs, until now, there are few theoretical results on synchronized stationary distribution of HSCSs although it is very meaningful in our real life. As a result, it constitutes the motivation of this paper. In this paper, our concern focuses on the existence of synchronized stationary distribution for HSCSs by studying the existence of stationary distribution of corresponding error system.

Recently, many useful methods have been proposed to study the existence of stationary distribution. In [18], Zou et al. presented the existence of stationary distribution for a noisy predator–prey system by Hasminskii’s method. Mao showed the existence of a unique stationary distribution for stochastic population systems by Lyapunov method and developed a useful method to compute the mean and variance of stationary distribution in [19]. In [20–22], Huang et al. investigated the existence, non-existence and degenerate diffusion of steady states for Fokker–Planck equations by using level set method and Lyapunov method, especially integral identity and measure estimates. Whereafter, based on the existence theory in [20], Liu et al. adopted graph-theoretical method to investigate the existence of stationary distribution of stochastic coupled systems (SCSs) in [23]. Furthermore, in [24], Zhu and Yin proposed a new method to investigate the existence of stationary distribution for stochastic hybrid systems. From then on, many researchers utilized this method to study stationary distribution for a variety of practical stochastic hybrid systems. For instance, in [25–27], the authors gained the existence of stationary distribution for stochastic ecological model and stochastic epidemic model through investigating ergodic property and positive recurrence. Likewise, drawing support from this method motivates our investigation in this paper.

Based on the existence theory of stationary distribution for stochastic hybrid systems in [24], we first investigate the existence of synchronized stationary distribution for HSCSs by leveraging Lyapunov method and graph theory [30], which averts the difficulty of directly constructing an appropriate Lyapunov function. Then, two kinds of sufficient criteria of synchronized stationary distribution of HSCSs with constant perturbation intensity are ob-

tained. What is exhibited is that stochastic perturbation intensity has a great influence on the existence region of synchronized stationary distribution. And synchronized stationary distribution will be replaced by complete synchronization, if stochastic perturbation is zero. The main contributions of this paper are listed as follows:

- (i) Based on the existence theory of stationary distribution in [24], a comprehensive synchronized stationary distribution analysis of HSCSs with constant perturbation intensity is investigated for the first time. As far as the authors know, there is little literature on this research.
- (ii) Graph theory is firstly applied to study the existence for synchronized stationary distribution of HSCSs, which simplifies some complex analyses and avoids some difficulties.
- (iii) The obtained theoretical results are utilized to tackle synchronized stationary distribution of stochastic coupled oscillators and a Chua's circuits network both with stochastic perturbation and Markovian switching, which reveals the broad applications of our theoretical results to various physical systems.

The rest of this paper is arranged as follows: Section 2 gives preliminaries and problem statement for HSCSs. In Section 3, sufficient conditions are derived for HSCSs, guaranteeing the existence of synchronized stationary distribution. Then, in Section 4, we apply the theoretical results to nonlinear stochastic coupled oscillators and a Chua's circuits network. The corresponding numerical simulations are also presented respectively. Finally, we show the conclusion in Section 5.

## 2. Preliminaries and problem statement

Throughout this paper, unless otherwise specified, we use the following notations. Let  $\mathbb{R}^n$  denote the  $n$ -dimensional Euclidean space. Write  $|\cdot|$  for the Euclidean norm of vectors or the trace norm of matrices. The superscript “ $T$ ” stands for the transpose of a vector or a matrix. Moreover, notations  $\mathbb{L} = \{1, 2, \dots, N\}$  and  $\mathbb{R}_+^1 = \{x \in \mathbb{R}^1 | x \geq 0\}$  are used. Denote by  $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, Pr)$  a complete probability space with a filtration  $\{\mathcal{F}_t\}_{t \geq 0}$  satisfying the usual conditions. Let  $r(t)$ ,  $t \geq 0$ , be a right-continuous Markov chain on the probability space taking values in a finite state space  $\mathbb{S} = \{1, 2, \dots, l\}$  with generator  $\mathfrak{T} = (\pi_{kh})_{l \times l}$  given by

$$Pr\{r(t + \Delta t) = h | r(t) = k\} = \begin{cases} \pi_{kh}\Delta t + o(\Delta t) & \text{if } h \neq k, \\ 1 + \pi_{kk}\Delta t + o(\Delta t) & \text{if } h = k, \end{cases}$$

where  $\pi_{kh} \geq 0$  is the transition rate from  $k$  to  $h$ , if  $k \neq h$ , while  $\pi_{kk} = -\sum_{h=1, h \neq k}^l \pi_{kh}$ . In this paper, we assume  $\pi_{kh} > 0$  for  $k \neq h$ . Let  $C^2(\mathbb{R}^m \times \mathbb{S}; \mathbb{R}_+^1)$  represent the family of twice continuously differentiable nonnegative real-valued functions  $v(x, k)$  defined on  $\mathbb{R}^m \times \mathbb{S}$ .

We introduce some useful concepts associated with a digraph. Let  $\mathcal{H} = \{\mathbb{L}, E\}$  be a digraph, where  $\mathbb{L}$  is the set of vertices and  $E$  is the set of arcs  $(i, j)$  leading from initial vertex  $i$  to terminal vertex  $j$  in digraph  $\mathcal{H}$ . A digraph is called strongly connected if for any two distinct vertices  $i$  and  $j$  in the digraph, there exists a path from vertex  $i$  to vertex  $j$ . For fixed  $k \in \mathbb{S}$ , define the weight matrix of  $\mathcal{H}$  as  $\Lambda(k) = (\eta_{ij}(k))_{N \times N}$ . And  $\eta_{ij}(k) > 0$ , if there exists an arc from vertex  $j$  to vertex  $i$ , and  $\eta_{ij}(k) = 0$  otherwise. The digraph  $\mathcal{H}$  with weight matrix  $\Lambda(k)$  is denoted by  $(\mathcal{H}, \Lambda(k))$ . The Laplacian matrix of  $\Lambda(k)$  is defined as  $L_p(\mathcal{H}) = (p_{ij}(k))_{N \times N}$ , where  $p_{ij}(k) = \sum_{r \neq i} \eta_{ir}(k)$  for  $i = j$  and  $p_{ij}(k) = -\eta_{ij}(k)$  for  $i \neq j$ . For more details on graph theory, we refer the reader to [31].

Here we consider the following HSCSs consisting of  $N(N \geq 2)$  identical coupled vertices with each vertex described by the following dynamical system:

$$dx_i(t) = \left[ f(x_i(t), r(t)) + \sum_{j=1}^N a_{ij}(r(t))H_{ij}(x_i(t) - x_j(t), r(t)) + I(r(t)) + u_i(x_i(t) - s(t), r(t)) \right] dt + \sigma_i(r(t))dB_i(t), \quad i \in \mathbb{L}, \quad (1)$$

where  $x_i(t) \in \mathbb{R}^m$  is the state vector associated with the  $i$ th vertex, function  $f(\cdot, \cdot) : \mathbb{R}^m \times \mathbb{S} \rightarrow \mathbb{R}^m$  is a continuous vector-valued function,  $H_{ij}(\cdot, \cdot) : \mathbb{R}^m \times \mathbb{S} \rightarrow \mathbb{R}^m$  represents the influence of the  $j$ th vertex on the  $i$ th vertex,  $a_{ij}(\cdot)$  indicates the coupling strength,  $I(\cdot) : \mathbb{S} \rightarrow \mathbb{R}^m$  is a constant external input vector, the term  $\sigma_i(\cdot)dB_i(t)$  represents the stochastic perturbation, in which  $\sigma_i(\cdot) = \text{diag}\{\sigma_i^{(1)}(\cdot), \sigma_i^{(2)}(\cdot), \dots, \sigma_i^{(m)}(\cdot)\}$ ,  $\sigma_i^{(r)} \in \mathbb{R}^1$  and  $\sigma_i^{(r)} \neq 0$ ,  $B_i(t) = (B_i^{(1)}(t), B_i^{(2)}(t), \dots, B_i^{(m)}(t))^T$  is an  $m$ -dimensional Brownian motion defined on  $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, Pr)$ . We assume that  $B_1(t), B_2(t), \dots, B_N(t)$  and  $r(t)$  are independent of each other.  $u_i(x_i(t) - s(t), r(t)) \in \mathbb{R}^m$  is the control input of the  $i$ th vertex, where  $s(t) \in \mathbb{R}^m$  is the state trajectory of an unforced isolate vertex satisfying

$$\dot{s}(t) = f(s(t), r(t)) + I(r(t)).$$

It can be an equilibrium point, a nontrivial periodic orbit, or even a chaotic orbit. Following the work in [30], we can describe system (1) on digraph  $\mathcal{H}$  with  $N$  vertices.

Let  $e_i(t) \triangleq x_i(t) - s(t)$  be the synchronization error. Then, the corresponding error system can be obtained as follows:

$$de_i(t) = \left[ F(e_i(t), r(t)) + u_i(e_i(t), r(t)) + \sum_{j=1}^N a_{ij}(r(t))H_{ij}(e_i(t) - e_j(t), r(t)) \right] dt + \sigma_i(r(t))dB_i(t), \quad i \in \mathbb{L}, \quad (2)$$

where  $F(e_i(t), r(t)) = f(x_i(t), r(t)) - f(s(t), r(t))$ .

For representation convenience, let  $e(t) = (e_1^T(t), e_2^T(t), \dots, e_N^T(t))^T \in \mathbb{R}^{mN}$ ,  $B(t) = (B_1^T(t), B_2^T(t), \dots, B_N^T(t))^T$ ,  $G(r(t)) = \text{diag}\{\sigma_1(r(t)), \sigma_2(r(t)), \dots, \sigma_N(r(t))\}$ ,

$$\bar{F}(e(t), r(t)) = \begin{pmatrix} F(e_1(t), r(t)) + u_1(e_1(t), r(t)) + \sum_{j=1}^N a_{1j}(r(t))H_{1j}(e_1(t) - e_j(t), r(t)) \\ \vdots \\ F(e_N(t), r(t)) + u_N(e_N(t), r(t)) + \sum_{j=1}^N a_{Nj}(r(t))H_{Nj}(e_N(t) - e_j(t), r(t)) \end{pmatrix} \in \mathbb{R}^{mN}.$$

Then system (2) can be rewritten as

$$de(t) = \bar{F}(e(t), r(t))dt + G(r(t))dB(t). \quad (3)$$

We denote  $e(0) = e_0$ ,  $r(0) = r_0$ ,  $C(r(t)) = G(r(t))G^T(r(t))$ .

In this paper, we assume that  $f(\cdot, \cdot)$ ,  $u_i(\cdot, \cdot)$  and  $H_{ij}(\cdot, \cdot)$  satisfy the usual local Lipschitz condition and the linear growth condition. Then, system (3) has a unique solution (Theorem 3.13 in [1]). It is claimed that synchronized stationary distribution of system (1) is achieved, if error dynamical system (2) exists a stationary distribution.

**Remark 1.** It is worth noting that this kind of coupling form we consider is common [16,29,32–35]. For example, in the Kuramoto oscillators on complex networks [33], the coupling function which represents the influence of oscillator  $i$  on oscillator  $j$  is  $\sin(\theta_i - \theta_j)$ ,



where  $\theta_i$  denotes the phase of the  $i$ th oscillator. And researchers usually consider two kinds of synchronization orbits in literature. One is that all points have the same trajectory  $s(t)$  (see [32,35]), and the other is that different points track different target orbits  $s_i(t)$  (see [36–38]). In this paper, we consider that all points have the same trajectory. However, in contrast to previous work, we focus on synchronized stationary distribution that addresses a weaker form of synchronization. For the latter, we reserve for the future research.

The following definition will assist in deriving our main results.

**Definition 1** [1]. For  $V_i(e_i, k) \in C^2(\mathbb{R}^m \times \mathbb{S}; \mathbb{R}_+^1)$ ,  $e_i = (e_i^{(1)}, e_i^{(2)}, \dots, e_i^{(m)})^T$ ,  $\mathcal{L}^*V_i(e_i, k)$  is called a differential operator expression associated with system (2) if

$$\begin{aligned} \mathcal{L}^*V_i(e_i, k) &= \sum_{h=1}^l \pi_{kh} V_i(e_i, h) + \frac{\partial V_i(e_i, k)}{\partial e_i} \left[ F(e_i(t), k) + u_i(e_i(t), k) + \sum_{j=1}^N a_{ij}(k) H_{ij}(e_i(t) - e_j(t), k) \right] \\ &\quad + \frac{1}{2} \text{Trace} \left\{ \sigma_i^T(k) \frac{\partial^2 V_i(e_i, k)}{\partial e_i^2} \sigma_i(k) \right\}, \end{aligned}$$

where

$$\frac{\partial V_i(e_i, k)}{\partial e_i} = \left( \frac{\partial V_i(e_i, k)}{\partial e_i^{(1)}}, \frac{\partial V_i(e_i, k)}{\partial e_i^{(2)}}, \dots, \frac{\partial V_i(e_i, k)}{\partial e_i^{(m)}} \right), \quad \frac{\partial^2 V_i(e_i, k)}{\partial e_i^2} = \left( \frac{\partial^2 V_i(e_i, k)}{\partial e_i^{(r)} \partial e_i^{(s)}} \right)_{m \times m}.$$

### 3. Main results

In this section, we shall focus our attention on the existence of synchronized stationary distribution for system (1). By resorting to Lemma 2.1 in [27], sufficient criteria of synchronized stationary distribution will be presented in the following.

**Theorem 1.** Let  $p \geq 2$ , if for any  $i \in \mathbb{L}$ ,  $k \in \mathbb{S}$ , there are constants  $\beta_i^{(k)}$ , nonnegative constants  $\eta_{ij}(k)$ ,  $\bar{O}_i(k)$  and functions  $M_j(e_j)$ ,  $j \in \mathbb{L}$ , satisfying

$$\mathcal{L}^*V_i(e_i, k) \leq -\beta_i^{(k)} |e_i|^p + \sum_{j=1}^N \eta_{ij}(k) (M_j(e_j) - M_i(e_i)) + \bar{O}_i(k), \quad (e_i, k) \in \mathcal{D}_i^c \times \mathbb{S}, \quad (4)$$

where  $\mathcal{D}_i^c = \overbrace{(-\rho, \rho)^c \times (-\rho, \rho)^c \times \dots \times (-\rho, \rho)^c}^m$ ,  $\rho > (\frac{\bar{O}_i(k)}{\beta_i^{(k)} m^{\frac{p}{2}}})^{\frac{1}{p}}$ , is a nonempty open set with compact closure, then system (1) admits a synchronized stationary distribution under the assumption that digraph  $(\mathcal{H}, \Lambda(k))$  is strongly connected, in which  $\Lambda(k) = (\eta_{ij}(k))_{N \times N}$ .

**Proof.** On the basis of the assumption in Section 2, we have  $\pi_{kh} > 0$  for  $k \neq h$ . Due to the fact that diffusion matrix  $C(k) = \text{diag}\{(\sigma_1^{(1)}(k))^2, (\sigma_1^{(2)}(k))^2, \dots, (\sigma_1^{(m)}(k))^2, \dots, (\sigma_N^{(1)}(k))^2, (\sigma_N^{(2)}(k))^2, \dots, (\sigma_N^{(m)}(k))^2\}$  is positive definite, it indicates that there exists some constant  $\lambda_1 \in (0, 1]$  such that

$$\lambda_1 |\zeta|^2 \leq \langle C(r(t)) \zeta, \zeta \rangle \leq \lambda_1^{-1} |\zeta|^2,$$

for all  $\zeta, e_0 \in \mathbb{R}^{mN}$ , where  $\langle \cdot, \cdot \rangle$  denotes inner product. Define

$$V(e, k) = \sum_{i=1}^N \gamma_i(k) V_i(e_i, k), \quad (e, k) \in \mathcal{D}^c \times \mathbb{S},$$

where  $\gamma_i(k)$  is the cofactor of the  $i$ th diagonal element of  $L_p(\mathcal{H})$  and  $\mathcal{D}^c \triangleq \overbrace{(-\rho, \rho)^c \times (-\rho, \rho)^c \times \cdots \times (-\rho, \rho)^c}^{mN}$  is a nonempty open set with compact closure. Because digraph  $(\mathcal{H}, \Lambda(k))$  is strongly connected, we have  $\gamma_i(k) > 0$  (see Proposition 2.1 in [30]). By Eq. (4), one can make the following calculation

$$\begin{aligned} \mathcal{L}^* V(e, k) &= \sum_{i=1}^N \gamma_i(k) \mathcal{L}^* V_i(e_i, k) \\ &\leq \sum_{i=1}^N \gamma_i(k) \left( -\beta_i^{(k)} |e_i|^p + \sum_{j=1}^N \eta_{ij}(k) (M_j(e_j) - M_i(e_i)) + \bar{O}_i(k) \right) \\ &= - \sum_{i=1}^N \gamma_i(k) \beta_i^{(k)} |e_i|^p + \sum_{i=1}^N \sum_{j=1}^N \gamma_i(k) \eta_{ij}(k) (M_j(e_j) - M_i(e_i)) + \sum_{i=1}^N \gamma_i(k) \bar{O}_i(k). \end{aligned} \quad (5)$$

By Theorem 2.2 in [30], one gets that

$$\sum_{i=1}^N \sum_{j=1}^N \gamma_i(k) \eta_{ij}(k) (M_j(e_j) - M_i(e_i)) = \sum_{\mathcal{Q} \in \mathbb{Q}} W(\mathcal{Q}) \sum_{(s,r) \in E(C_{\mathcal{Q}})} (M_r(e_r) - M_s(e_s)), \quad (6)$$

where  $\mathbb{Q}$  is the set of all spanning unicyclic graphs of  $\mathcal{H}$ ,  $W(\mathcal{Q})$  is the weight of  $\mathcal{Q}$ , and  $C_{\mathcal{Q}}$  denotes the directed cycle of  $\mathcal{Q}$ . Without loss of generality, for any directed cycle  $C_{\mathcal{Q}}$ , the set  $E(C_{\mathcal{Q}})$  can be described as

$$E(C_{\mathcal{Q}}) = \{(i_k, i_{k+1}) | k = 1, 2, \dots, n-1, n \leq N, i_{n+1} = i_1\}. \quad (7)$$

Based on Eq. (7), we have

$$\begin{aligned} \sum_{(s,r) \in E(C_{\mathcal{Q}})} (M_r(e_r) - M_s(e_s)) &= M_1(e_1) - M_2(e_2) + M_2(e_2) - M_3(e_3) + \cdots + M_{n-1}(e_{n-1}) \\ &\quad - M_n(e_n) + M_n(e_n) - M_1(e_1) = 0. \end{aligned} \quad (8)$$

It can be checked from Eqs. (5) to (8) that:

$$\mathcal{L}^* V(e, k) \leq - \sum_{i=1}^N \gamma_i(k) \beta_i^{(k)} |e_i|^p + \sum_{i=1}^N \gamma_i(k) \bar{O}_i(k).$$

For any  $e_i^{(r)} \in (-\rho, \rho)^c$ , we get that  $|e_i|^p \geq m^{\frac{p}{2}} \rho^p$ . Thus,

$$\begin{aligned}\mathcal{L}^*V(e, k) &\leq -\sum_{i=1}^N \gamma_i(k) \beta_i^{(k)} m^{\frac{p}{2}} \rho^p + \sum_{i=1}^N \gamma_i(k) \bar{O}_i(k) \\ &= -\sum_{i=1}^N \gamma_i(k) \left( \beta_i^{(k)} m^{\frac{p}{2}} \rho^p - \bar{O}_i(k) \right).\end{aligned}$$

We can arrive at that

$$\mathcal{L}^*V(e, k) \leq -\varepsilon, \quad (e, k) \in \mathcal{D}^c \times \mathbb{S},$$

where  $\varepsilon = \min_{k \in \mathbb{S}} \{ \sum_{i=1}^N \gamma_i(k) (\beta_i^{(k)} m^{\frac{p}{2}} \rho^p - \bar{O}_i(k)) \} > 0$  due to  $\beta_i^{(k)} m^{\frac{p}{2}} \rho^p - \bar{O}_i(k) > 0$  by  $\rho > (\bar{O}_i(k) / (\beta_i^{(k)} m^{\frac{p}{2}}))^{\frac{1}{p}}$ . Therefore, there exists a unique stationary distribution of system (2) according to Lemma 2.1 in [27], which means that system (1) admits a synchronized stationary distribution. The proof is completed.  $\square$

**Remark 2.** We construct a global Lyapunov function as  $V(e, k) = \sum_{i=1}^N \gamma_i(k) V_i(e_i, k)$ , which indicates that synchronized stationary distribution of system (1) is related to the topological structure of the coupled network. Meanwhile,  $\gamma_i(k)$  can assist us to deal with coupling term effectively, which makes us avoid using Kronecker product method and linear matrix inequality. This construction was first proposed by Li and Shuai [30], which has been successfully applied to studying the dynamic behaviors of large-scale systems, including stability [9,39] and synchronization [12,13]. As one of essential dynamic behaviors, stationary distribution of SCSs was investigated by Liu et al. [23] through adopting the combination of graph theory and Lyapunov method. Based on the research of Liu et al., we extend graph theory to synchronized stationary distribution analysis of HSCSs for the first time.

The conditions in Theorem 1 are general (see [9,11–13]). However, since functions  $M_i(e_i)$ ,  $V_i(e_i, k)$  are unknown, Theorem 1 may lead to conservativeness and inconvenience in real applications. On the other hand, from Theorem 1, we can see that  $\varepsilon$  and  $\rho$  affect synchronized stationary distribution. In fact,  $\varepsilon$  and  $\rho$  are essentially determined by perturbation intensity and control intensity, which is not intuitively reflected in Theorem 1. Thus, on the basis of Theorem 1, we further present the following theorem whose conditions depend on the coefficients of the error system.

**Theorem 2.** Let  $p \geq 2$ . Assume that the following conditions are satisfied.

(i) For any  $x_i, s \in \mathbb{R}^m$ , there exist constants  $\alpha^{(k)}, \epsilon_i^{(k)}$  and positive constant  $P_{ij}(k)$  such that

$$e_i^T (f(x_i, k) - f(s, k)) \leq \alpha^{(k)} |x_i - s|^2, \quad e_i^T u_i(e_i, k) \leq \epsilon_i^{(k)} |e_i|^2, \quad (9)$$

and

$$|H_{ij}(x_i, k)| \leq P_{ij}(k) |x_i|, \quad i, j \in \mathbb{L}, k \in \mathbb{S}. \quad (10)$$

(ii) For each  $i \in \mathbb{L}$ , matrix

$$\begin{aligned}\mathcal{B}_i &:= -\text{diag} \left\{ p(\alpha^{(1)} + \epsilon_i^{(1)}) + \frac{1}{2}(p-1)(p-2) + D_i^{(1)}, \dots, p(\alpha^{(l)} + \epsilon_i^{(l)}) \right. \\ &\quad \left. + \frac{1}{2}(p-1)(p-2) + D_i^{(l)} \right\} - \mathfrak{T},\end{aligned}$$

is a nonsingular  $M$ -matrix, where  $D_i^{(k)} = \sum_{j=1}^N \frac{1}{2} p a_{ij}(k) + \sum_{j=1}^N 2 p a_{ij}(k) P_{ij}^2(k)$ .  
 (iii) Digraph  $(\mathcal{H}, (a_{ij}(k))_{N \times N})$  is strongly connected.

Then system (1) admits a synchronized stationary distribution.

**Proof.** Since  $\mathcal{B}_i$  is a nonsingular  $M$ -matrix, we can find  $q_i^{(k)} > 0$  and  $\beta_i^{(k)} > 0$  such that  $\beta_i = \mathcal{B}_i q_i$ , where  $\beta_i = (\beta_i^{(1)}, \beta_i^{(2)}, \dots, \beta_i^{(l)})^T$  and  $q_i = (q_i^{(1)}, q_i^{(2)}, \dots, q_i^{(l)})^T$ . For any  $i \in \mathbb{L}$ ,  $k \in \mathbb{S}$ , we define

$$V_i(e_i, k) = q_i^{(k)} |e_i|^p, \quad (e_i, k) \in \mathcal{D}_i^c \times \mathbb{S},$$

where  $\mathcal{D}_i^c = \overbrace{(-\rho, \rho)^c \times (-\rho, \rho)^c \times \dots \times (-\rho, \rho)^c}^m$  is a nonempty open set with compact closure in which  $\rho > (\frac{\bar{O}_i(k)}{\beta_i^{(k)} m^{\frac{p}{2}}})^{\frac{1}{p}}$ ,  $\bar{O}_i(k) = (p-1) q_i^{(k)} |\sigma_i(k)|^p$ . Hence, according to the  $\mathcal{L}^*$  operator in Definition 1 and Eqs. (9) and (10), we have

$$\begin{aligned} \mathcal{L}^* V_i(e_i, k) &= \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p |e_i|^{p-2} e_i^T \left[ F(e_i, k) + u_i(e_i, k) + \sum_{j=1}^N a_{ij}(k) H_{ij}(e_i - e_j, k) \right] \\ &\quad + \frac{1}{2} \text{Trace} \left\{ \sigma_i^T(k) q_i^{(k)} (p |e_i|^{p-2} I + p(p-2) |e_i|^{p-4} e_i e_i^T) \sigma_i(k) \right\} \\ &\leq \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p \alpha^{(k)} |e_i|^p + q_i^{(k)} p \epsilon_i^{(k)} |e_i|^p \\ &\quad + q_i^{(k)} p |e_i|^{p-2} \sum_{j=1}^N e_i^T a_{ij}(k) H_{ij}(e_i - e_j, k) + \frac{1}{2} p(p-1) |\sigma_i(k)|^2 q_i^{(k)} |e_i|^{p-2} \\ &\leq \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p \alpha^{(k)} |e_i|^p + q_i^{(k)} p \epsilon_i^{(k)} |e_i|^p \\ &\quad + \frac{1}{2} q_i^{(k)} p |e_i|^{p-2} \sum_{j=1}^N a_{ij}(k) (|e_i|^2 + |H_{ij}(e_i - e_j, k)|^2) \\ &\quad + \frac{1}{2} p(p-1) |\sigma_i(k)|^2 q_i^{(k)} |e_i|^{p-2} \\ &\leq \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p \alpha^{(k)} |e_i|^p + q_i^{(k)} p \epsilon_i^{(k)} |e_i|^p + \frac{1}{2} q_i^{(k)} p \sum_{j=1}^N a_{ij}(k) |e_i|^p \\ &\quad + \frac{1}{2} q_i^{(k)} p |e_i|^{p-2} \sum_{j=1}^N a_{ij}(k) P_{ij}^2(k) |e_i - e_j|^2 + \frac{1}{2} p(p-1) |\sigma_i(k)|^2 q_i^{(k)} |e_i|^{p-2} \\ &\leq \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p \alpha^{(k)} |e_i|^p + q_i^{(k)} p \epsilon_i^{(k)} |e_i|^p + \frac{1}{2} q_i^{(k)} p \sum_{j=1}^N a_{ij}(k) |e_i|^p \end{aligned}$$

$$\begin{aligned}
& + \sum_{j=1}^N p q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^p + \sum_{j=1}^N p q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^{p-2} |e_j|^2 \\
& + \frac{1}{2} p(p-1) |\sigma_i(k)|^2 q_i^{(k)} |e_i|^{p-2}.
\end{aligned} \tag{11}$$

Based on Young's inequality, we can get that

$$\begin{aligned}
\sum_{j=1}^N p q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^{p-2} |e_j|^2 & \leq \sum_{j=1}^N (p-2) q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^p + \sum_{j=1}^N 2 q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_j|^p, \\
\frac{1}{2} p(p-1) q_i^{(k)} |\sigma_i(k)|^2 |e_i|^{p-2} & \leq \frac{1}{2} (p-1)(p-2) q_i^{(k)} |e_i|^p + (p-1) q_i^{(k)} |\sigma_i(k)|^p.
\end{aligned}$$

Therefore, we can derive

$$\begin{aligned}
\mathcal{L}^* V_i(e_i, k) & \leq \sum_{h=1}^l \pi_{kh} q_i^{(h)} |e_i|^p + q_i^{(k)} p \alpha^{(k)} |e_i|^p + q_i^{(k)} p \epsilon_i^{(k)} |e_i|^p + \frac{1}{2} q_i^{(k)} p \sum_{j=1}^N a_{ij}(k) |e_i|^p \\
& + \sum_{j=1}^N p q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^p + \sum_{j=1}^N (p-2) q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_i|^p \\
& + \sum_{j=1}^N 2 q_i^{(k)} a_{ij}(k) P_{ij}^2(k) |e_j|^p + \frac{1}{2} (p-1)(p-2) q_i^{(k)} |e_i|^p + (p-1) q_i^{(k)} |\sigma_i(k)|^p \\
& = \left[ \sum_{h=1}^l \pi_{kh} q_i^{(h)} + q_i^{(k)} p (\alpha^{(k)} + \epsilon_i^{(k)}) + \frac{1}{2} q_i^{(k)} p \sum_{j=1}^N a_{ij}(k) + \sum_{j=1}^N 2 p q_i^{(k)} a_{ij}(k) P_{ij}^2(k) \right. \\
& \quad \left. + \frac{1}{2} (p-1)(p-2) q_i^{(k)} \right] |e_i|^p + \sum_{j=1}^N 2 q_i^{(k)} a_{ij}(k) P_{ij}^2(k) (|e_j|^p - |e_i|^p) \\
& \quad + (p-1) q_i^{(k)} |\sigma_i(k)|^p \\
& = -\beta_i^{(k)} |e_i|^p + \sum_{j=1}^N \eta_{ij}(k) (M_j(e_j) - M_i(e_i)) + \bar{O}_i(k),
\end{aligned}$$

where  $\eta_{ij}(k) = 2 q_i^{(k)} a_{ij}(k) P_{ij}^2(k)$ ,  $M_j(e_j) = |e_j|^p$ . It implies that Eq. (4) holds. Since digraph  $(\mathcal{H}, (a_{ij}(k))_{N \times N})$  is strongly connected, then digraph  $(\mathcal{H}, (\eta_{ij}(k))_{N \times N})$  is also strongly connected. Therefore, it follows from Theorem 1 that system (1) admits a synchronized stationary distribution. This completes the proof.  $\square$

**Remark 3.** It should be noticed that the existence region of synchronized stationary distribution is closely related to stochastic perturbation intensity and feedback control intensity. Large perturbation intensity and small control intensity can achieve greater synchronized stationary distribution region. It's worth noting that if we let  $\sigma_i(r(t)) \equiv 0$ , and Eq. (4) in Theorem 1 is replaced by the following inequality

$$\mathcal{L}^* V_i(e_i, k) \leq -\beta_i^{(k)} |e_i|^p + \sum_{j=1}^N \eta_{ij}(k) (M_j(e_j) - M_i(e_i)),$$

for all  $(e_i, k)$ . This means that  $\mathcal{L}^*V(e, k) \leq 0$  holds without restriction on the domain, which shows system (1) achieves complete synchronization. These will be explained in the last numerical simulation of this paper.

**Remark 4.** Up to now, many results about stationary distribution of stochastic systems have been obtained in [20–22,24,27]. It is worth noting that the results obtained in the above literatures do not consider the coupling factors. And compared with literature [20–23], our model considers both coupling factor and Markovian switching, which is more general. In [29], the authors taken coupling into consideration, which investigated synchronized stationary states of the Kuramoto model with a specific coupling form. Meanwhile, we study a special stationary distribution, synchronized stationary distribution, of stochastic coupled systems, which is different from [23]. In comparison with references [18,19,25–28], in our paper, we study the synchronized stationary distribution of stochastic coupled oscillators and Chua's circuit network. The above literature all considered biological models, such as population models, epidemic models and so on. Therefore, our research extends the practical application in physics of stationary distribution.

## 4. Applications

In order to verify the performance of the proposed theoretical results, two typical stochastic hybrid systems, including nonlinear stochastic coupled oscillators and a Chua's circuits network, are taken as applications of our main results.

### 4.1. Synchronized stationary distribution analysis for stochastic coupled oscillators

In this subsection, we will apply main results to discussing the existence of synchronized stationary distribution for a stochastic coupled oscillators model. Nonlinear oscillator

$$\ddot{x}(t) + \varphi\dot{x}(t) + g(x(t)) = 0, \quad t \geq 0, \quad (12)$$

is widely used in many fields. Many researchers have recently investigated the dynamic properties of system (12), including stability, synchronization, boundedness and so on, but little literature study synchronized stationary distribution.

Here we shall consider nonlinear coupled oscillators with Markovian switching described by

$$\begin{aligned} \ddot{x}_i(t) + \varphi(r(t))\dot{x}_i(t) + x_i(t) + g(x_i(t), r(t)) + \sum_{j=1}^N c_{ij}(r(t))\tilde{G}_{ij}(x_i(t) - x_j(t), r(t)) \\ = I(r(t)), \quad i \in \mathbb{L}, \end{aligned} \quad (13)$$

where  $x_i(t) \in \mathbb{R}^1$  is the state variable,  $g(\cdot, \cdot) : \mathbb{R}^1 \times \mathbb{S} \rightarrow \mathbb{R}^1$  is a continuous nonlinear function,  $c_{ij}(\cdot)$  is the coupling strength, function  $\tilde{G}_{ij}(\cdot, \cdot) : \mathbb{R}^1 \times \mathbb{S} \rightarrow \mathbb{R}^1$  is the coupling form and  $\varphi(\cdot) \geq 0$  represents the damping coefficient.

Let  $y_i(t) = \dot{x}_i(t) + \xi x_i(t)$ , in which  $\xi$  is a positive constant. Then, taking the influence of control input  $u_i^{(1)}$ ,  $u_i^{(2)}$  and stochastic perturbation  $\sigma_i^{(1)}(r(t))dB_i^{(1)}(t)$ ,  $\sigma_i^{(2)}(r(t))dB_i^{(2)}(t)$  into

account, system (13) can be rewritten as the following form:

$$\begin{cases} dx_i(t) = [y_i(t) - \xi x_i(t) + u_i^{(1)}(x_i(t) - s(t), r(t))]dt + \sigma_i^{(1)}(r(t))dB_i^{(1)}(t), \\ dy_i(t) = [(\xi - \varphi(r(t)))y_i(t) + (\xi\varphi(r(t)) - \xi^2 - 1)x_i(t) + u_i^{(2)}(y_i(t) - \bar{s}(t), r(t)) - g(x_i(t), r(t)) \\ - \sum_{j=1}^N c_{ij}(r(t))\tilde{G}_{ij}(x_i(t) - x_j(t), r(t)) + I(r(t))]dt + \sigma_i^{(2)}(r(t))dB_i^{(2)}(t), \quad i \in \mathbb{L}, \end{cases} \quad (14)$$

where  $\sigma_i^{(1)}(r(t)) \neq 0$ ,  $\sigma_i^{(2)}(r(t)) \neq 0$ ,  $s(t), \bar{s}(t) \in \mathbb{R}^1$  satisfy

$$\begin{cases} ds(t) = [\bar{s}(t) - \xi s(t)]dt, \\ d\bar{s}(t) = [(\xi - \varphi(r(t)))\bar{s}(t) + (\xi\varphi(r(t)) - \xi^2 - 1)s(t) - g(s(t), r(t)) + I(r(t))]dt. \end{cases} \quad (15)$$

In fact, system (15) is derived by the state trajectory of an unforced isolate vertex satisfying

$$\dot{s}(t) + \varphi(r(t))\dot{s}(t) + s(t) + g(s(t), r(t)) = I(r(t)) \quad (16)$$

and  $\bar{s}(t) = \dot{s}(t) + \xi s(t)$ . System (14) can be constructed on digraph  $\mathcal{H}$  with  $N$  vertices.

Denote  $e_{i1}(t) \triangleq x_i(t) - s(t)$ ,  $e_{i2}(t) \triangleq y_i(t) - \bar{s}(t)$ , which can lead to the following error system

$$\begin{cases} de_{i1}(t) = [e_{i2}(t) - \xi e_{i1}(t) + u_i^{(1)}(e_{i1}(t), r(t))]dt + \sigma_i^{(1)}(r(t))dB_i^{(1)}(t), \\ de_{i2}(t) = [(\xi - \varphi(r(t)))e_{i2}(t) + (\xi\varphi(r(t)) - \xi^2 - 1)e_{i1}(t) + u_i^{(2)}(e_{i2}(t), r(t)) - \bar{g}(e_{i1}(t), r(t)) \\ - \sum_{j=1}^N c_{ij}(r(t))\tilde{G}_{ij}(e_{i1}(t) - e_{j1}(t), r(t))]dt + \sigma_i^{(2)}(r(t))dB_i^{(2)}(t), \quad i \in \mathbb{L}, \end{cases} \quad (17)$$

where  $\bar{g}(e_{i1}(t), r(t)) = g(x_i(t), r(t)) - g(s(t), r(t))$ .

Here, the following assumptions are introduced for the technical reasons.

( $H_1$ ) For any  $x, s \in \mathbb{R}^1$ , there exist positive constants  $\mathfrak{H}(k)$  such that

$$|g(x, k) - g(s, k)| \leq \vartheta(k)|x - s|, \quad k \in \mathbb{S}, \quad (18)$$

with  $2\varphi(k) < 3 + 2\xi$ ,  $\varphi(k) > 2\xi$ ,  $\vartheta(k) \leq \frac{\xi}{4}$ .

( $H_2$ ) For any  $x \in \mathbb{R}^1$ , suppose that there are positive constants  $Z_{ij}(k)$  satisfying

$$|\tilde{G}_{ij}(x, k)| \leq Z_{ij}(k)|x|, \quad i, j \in \mathbb{L}, k \in \mathbb{S}.$$

**Remark 5.** Eq. (18) in assumption ( $H_1$ ) is typical in the study of SCSs (see [9,11–13]). For example, all linear and piecewise-linear time-invariant continuous functions satisfy this condition. Especially, the condition is satisfied if  $\frac{\partial g(x,k)}{\partial x}$  is uniformly bounded [35]. Similar conclusions also hold for assumption ( $H_2$ ).

Next, we give the following theorem which can assure the existence of synchronized stationary distribution for stochastic coupled oscillators with Markovian switching (14).

**Theorem 3.** Let  $p \geq 2$ , for each  $i \in \mathbb{L}, k \in \mathbb{S}$ , if matrix

$$\mathcal{B}_i := -\text{diag}\left\{p\Psi_i^{(1)} + \frac{1}{2}(p-1)(p-2) + D_i^{(1)}, \dots, p\Psi_i^{(l)} + \frac{1}{2}(p-1)(p-2) + D_i^{(l)}\right\} - \mathfrak{T}$$



is a nonsingular  $M$ -matrix provided assumptions  $(H_1)$  and  $(H_2)$ , where

$$\Psi_i^{(k)} = -\frac{\xi}{8} + \max \left\{ \theta_i^{(k)}, \iota_i^{(k)} \right\}, \quad D_i^{(k)} = \sum_{j=1}^N \frac{1}{2} p c_{ij}(k) + \sum_{j=1}^N 2 p c_{ij}(k) Z_{ij}^2(k),$$

and digraph  $(\mathcal{H}, (c_{ij}(k))_{N \times N})$  is strongly connected, then system (14) admits a synchronized stationary distribution.

**Proof.** For the following analysis, let  $e_i = (e_{i1}, e_{i2})^T$ ,  $\bar{G}_i(r(t)) = \text{diag}\{\sigma_i^{(1)}(r(t)), \sigma_i^{(2)}(r(t))\}$ ,  $B_i = (B_i^{(1)}, B_i^{(2)})^T$ ,

$$F_i(e_i, r(t)) = \begin{pmatrix} e_{i2} - \xi e_{i1} \\ (\xi - \varphi(r(t)))e_{i2} + (\xi \varphi(r(t)) - \xi^2 - 1)e_{i1} - \bar{g}(e_{i1}, r(t)) \end{pmatrix},$$

$$G_{ij}(e_i - e_j, r(t)) = \begin{pmatrix} 0 \\ -\tilde{G}_{ij}(e_{i1} - e_{j1}, r(t)) \end{pmatrix},$$

$$\bar{u}_i(e_i, r(t)) = \begin{pmatrix} u_i^{(1)}(e_{i1}, r(t)) \\ u_i^{(2)}(e_{i2}, r(t)) \end{pmatrix} = \begin{pmatrix} \theta_i^{(r(t))} e_{i1} \\ \iota_i^{(r(t))} e_{i2} \end{pmatrix}.$$

Then system (17) becomes

$$de_i(t) = \left[ F_i(e_i(t), r(t)) + \bar{u}_i(e_i(t), r(t)) + \sum_{j=1}^N c_{ij}(r(t)) G_{ij}(e_i(t) - e_j(t), r(t)) \right] dt \\ + \bar{G}_i(r(t)) dB_i(t), \quad i \in \mathbb{L}.$$

By simple calculation, we have

$$e_i^T F_i(e_i, k) = e_{i1} e_{i2} - \xi e_{i1}^2 + (\xi - \varphi(k)) e_{i2}^2 + (\xi \varphi(k) - \xi^2 - 1) e_{i1} e_{i2} - e_{i2} (g(x_i, k) - g(s, k)) \\ = -\xi |e_{i1}|^2 + (\xi - \varphi(k)) |e_{i2}|^2 + (\xi \varphi(k) - \xi^2) e_{i1} e_{i2} - e_{i2} (g(x_i, k) - g(s, k)) \\ \leq -\xi |e_{i1}|^2 + (\xi - \varphi(k)) |e_{i2}|^2 + \frac{\xi(\varphi(k) - \xi)}{2} \left( \varepsilon^2(k) |e_{i1}|^2 + \frac{|e_{i2}|^2}{\varepsilon^2(k)} \right) \\ + \frac{1}{2} \left( \delta^2(k) |e_{i2}|^2 + \frac{|g(x_i, k) - g(s, k)|^2}{\delta^2(k)} \right). \quad (19)$$

Let  $\varepsilon^2(k) = \frac{3}{2(\varphi(k) - \xi)}$  and  $\delta^2(k) = \frac{\xi}{4}$  for all  $k \in \mathbb{S}$ . Substituting these expressions into Eq. (19), and according to assumption  $(H_1)$ , it leads to

$$e_i^T F_i(e_i, k) \leq -\xi |e_{i1}|^2 - \xi |e_{i2}|^2 + \frac{3}{4} \xi |e_{i1}|^2 + \frac{\xi(\varphi(k) - \xi)^2}{3} |e_{i2}|^2 + \frac{\xi}{8} |e_{i2}|^2 \\ + \frac{2}{\xi} |g(x_i, k) - g(s, k)|^2 \\ \leq -\xi |e_{i1}|^2 - \xi |e_{i2}|^2 + \frac{3}{4} \xi |e_{i1}|^2 + \frac{3}{4} \xi |e_{i2}|^2 + \frac{\xi}{8} |e_{i2}|^2 + \frac{2}{\xi} \vartheta^2(k) |e_{i1}|^2 \\ \leq -\xi |e_{i1}|^2 - \xi |e_{i2}|^2 + \frac{3}{4} \xi |e_{i1}|^2 + \frac{3}{4} \xi |e_{i2}|^2 + \frac{\xi}{8} |e_{i2}|^2 + \frac{\xi}{8} |e_{i1}|^2 \\ = -\frac{\xi}{8} (|e_{i1}|^2 + |e_{i2}|^2) = -\frac{\xi}{8} |e_i|^2.$$

Table 1  
Related parameters.

| Parameters | $D_i^{(1)}$ | $D_i^{(2)}$ | $\Psi_i^{(1)}$ | $\Psi_i^{(2)}$ | $\sigma_i^{(1)}(1)$ | $\sigma_i^{(1)}(2)$ | $\sigma_i^{(2)}(1)$ | $\sigma_i^{(2)}(2)$ |
|------------|-------------|-------------|----------------|----------------|---------------------|---------------------|---------------------|---------------------|
| $i = 1$    | 0.08        | 0.073       | -0.428         | -0.328         | 4.15                | 2.15                | 4.15                | 2.15                |
| $i = 2$    | 0.04        | 0.146       | -0.446         | -0.328         | 4.25                | 2.25                | 4.25                | 2.25                |
| $i = 3$    | 0.04        | 0.024       | -0.446         | -0.328         | 4.35                | 2.35                | 4.35                | 2.35                |
| $i = 4$    | 0.06        | 0.171       | -0.428         | -0.328         | 4.45                | 2.45                | 4.45                | 2.45                |

In the meantime, one can arrive at that

$$e_i^T \bar{u}_i(e_i, k) = \theta_i^{(k)} |e_{i1}|^2 + \iota_i^{(k)} |e_{i2}|^2 \leq \max \left\{ \theta_i^{(k)}, \iota_i^{(k)} \right\} |e_i|^2.$$

Therefore, we can get that Eq. (9) in Theorem 2 is satisfied. In view of assumption  $(H_2)$ , we clearly have that

$$|G_{ij}(e_i - e_j, k)| = |\tilde{G}_{ij}(e_{i1} - e_{j1}, k)| \leq Z_{ij}(k) |e_{i1} - e_{j1}| \leq Z_{ij}(k) |e_i - e_j|.$$

Then, Eq. (10) holds. Furthermore, since  $\mathcal{B}_i$  is a nonsingular M-matrix, until now, all the conditions of Theorem 2 have been checked. As a result, system (14) can achieve synchronized stationary distribution. The proof is completed.  $\square$

#### 4.2. A numerical simulation for stochastic coupled oscillators

Next, we give a numerical simulation to demonstrate the effectiveness of the above-mentioned theoretical results.

Consider stochastic coupled oscillators with Markovian switching in the form of system (14). Let  $N = 4$ ,  $r(t)$  is a right continuous Markov chain taking values in  $\mathbb{S} = \{1, 2\}$  with generator

$$\mathfrak{T} = (\pi_{kh})_{2 \times 2} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

We choose  $\xi = 1.42$ ,  $\varphi(1) = 2.85$ ,  $\varphi(2) = 2.9$ . Meanwhile, the functions are chosen as follows:

$$g(x_i, 1) = 0.35 \sin x_i, \quad g(x_i, 2) = 0.34 \sin x_i, \quad i = 1, 2, 3, 4,$$

$$G_{ij}(x_i - x_j, 1) = 0.3(x_i - x_j), \quad G_{ij}(x_i - x_j, 2) = 0.4(x_i - x_j), \quad i, j = 1, 2, 3, 4.$$

Set  $\vartheta(1) = 0.352$ ,  $\vartheta(2) = 0.343$ ,  $Z_{ij}(1) = 0.5$ ,  $Z_{ij}(2) = 0.6$ ,  $i, j = 1, 2, 3, 4$ . We can find that assumptions  $(H_1)$  and  $(H_2)$  hold. Then, define

$$(c_{ij}(1))_{N \times N} = \begin{pmatrix} 0 & 0 & 0.02 & 0.02 \\ 0.01 & 0 & 0.01 & 0 \\ 0 & 0 & 0 & 0.02 \\ 0 & 0.03 & 0 & 0 \end{pmatrix}, \quad (c_{ij}(2))_{N \times N} = \begin{pmatrix} 0 & 0 & 0.03 & 0 \\ 0.02 & 0 & 0 & 0.04 \\ 0 & 0.01 & 0 & 0 \\ 0 & 0.04 & 0.03 & 0 \end{pmatrix}.$$

Obviously, digraph  $(\mathcal{H}, (c_{ij}(k))_{N \times N})$ ,  $k = 1, 2$  are strongly connected. Let control coefficients be as follows:  $\theta_i^{(1)} = -0.25$ ,  $\theta_i^{(2)} = -0.15$ ,  $\iota_i^{(1)} = -0.35$ ,  $\iota_i^{(2)} = -0.30$ ,  $i = 1, 2, 3, 4$ . Set  $p = 2$ . And the remaining parameters are listed in Table 1.

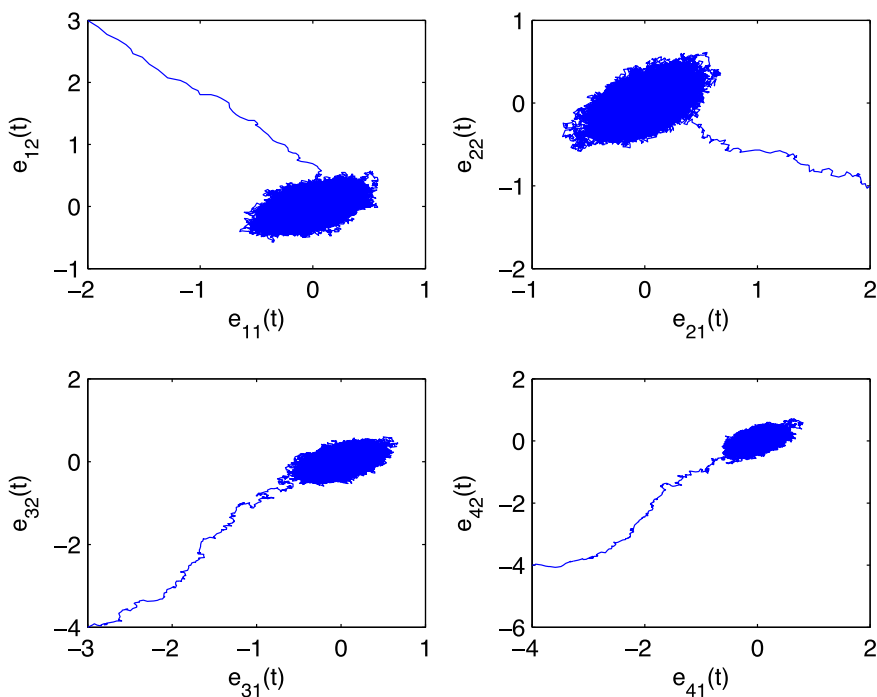


Fig. 1. The distribution of sample path of system (17).

Therefore, we can obtain that

$$\mathcal{B}_1 = -\text{diag}\{-0.775, -0.582\} - \mathfrak{T} = \begin{pmatrix} 1.775 & -1 \\ -1 & 1.582 \end{pmatrix},$$

$$\mathcal{B}_2 = -\text{diag}\{-0.851, -0.509\} - \mathfrak{T} = \begin{pmatrix} 1.851 & -1 \\ -1 & 1.509 \end{pmatrix},$$

$$\mathcal{B}_3 = -\text{diag}\{-0.851, -0.631\} - \mathfrak{T} = \begin{pmatrix} 1.851 & -1 \\ -1 & 1.631 \end{pmatrix},$$

$$\mathcal{B}_4 = -\text{diag}\{-0.795, -0.484\} - \mathfrak{T} = \begin{pmatrix} 1.795 & -1 \\ -1 & 1.484 \end{pmatrix}.$$

They are nonsingular M-matrices distinctly. In summary, all the conditions in Theorem 3 are verified. Then system (14) admits a synchronized stationary distribution. Furthermore, the initial values are selected as follows:  $e_0 = (-3, 4, 1, 0, -4, -3, -5, -3)^T$ . We can get the figures of the distribution of sample path in the phase space in Fig. 1 and the corresponding stationary distribution with the initial values of system (17) in Fig. 2. From Figs. 1 and 2, we can see that system (14) admits a synchronized stationary distribution. These numerical results show the effectiveness of our theoretical results.

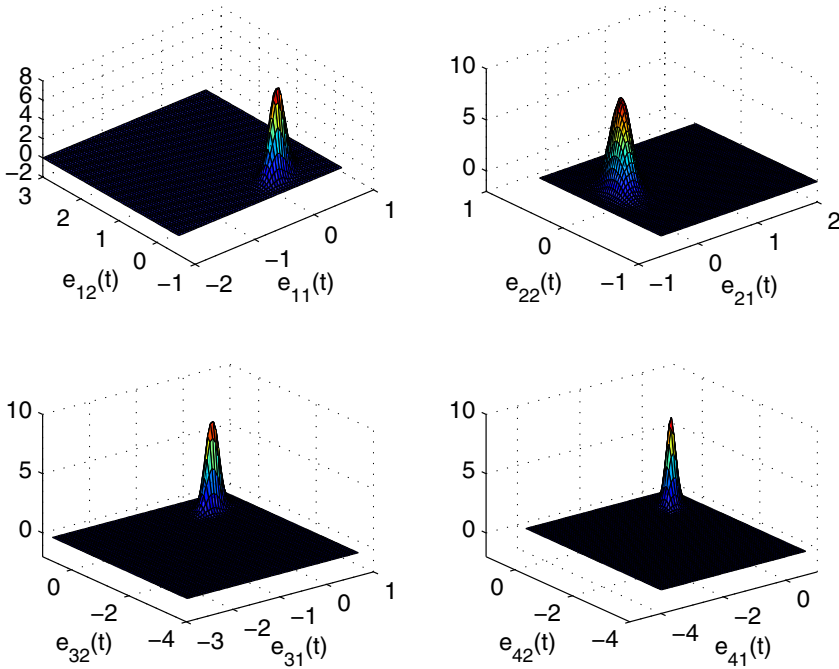


Fig. 2. The stationary distribution of system (17) with the given initial values.

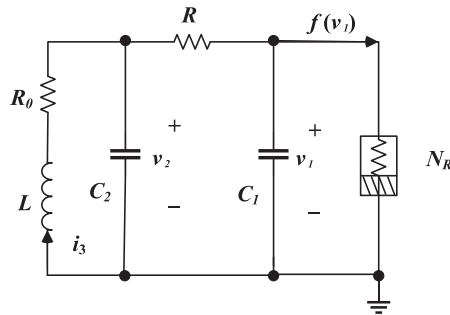


Fig. 3. The single Chua's circuit.

#### 4.3. Synchronized stationary distribution analysis for a Chua's circuits network

The well-known nonlinear electronic model of Chua's circuit is widely recognized as a robust, simple and economical electronic implementation of a nonlinear system exhibiting complex dynamics. It has been extensively studied from a theoretical, numerical and experimental point of view [40]. The single Chua's circuit depicted in Fig. 3 is described by the following differential equations (see Ref. [41]):

$$\begin{aligned} C_1 \dot{v}_1(t) &= \frac{1}{R}(-v_1(t) + v_2(t)) - f(v_1(t)), & C_2 \dot{v}_2(t) &= \frac{1}{R}(v_1(t) - v_2(t)) + i_3(t), \\ L \dot{i}_3(t) &= -(v_2(t) + R_0 i_3(t)), \end{aligned}$$

where  $v_1$  and  $v_2$  are the voltages across the capacitors  $C_1$  and  $C_2$ , respectively,  $i_3$  denotes the current through the inductances  $L$ ,  $R_0$  and  $R$  are the linear resistors, and the term  $f(v_1)$  represents the current through the nonlinear resistor  $N_R$ , which is a piecewise-linear function expressed as  $f(v_1) = G_{b1}v_1 + 0.5(G_{a1} - G_{b1})(|v_1 + 1| - |v_1 - 1|)$ . In general,  $G_{a1}$  and  $G_{b1}$  are negative constants.

Consider a network of  $N(N \geq 2)$  Chua's circuits with coupling term, the networked system with Markovian switching, stochastic perturbation and control input is defined as follows:

$$\begin{aligned} \begin{pmatrix} \dot{x}_{i1} \\ \dot{x}_{i2} \\ \dot{x}_{i3} \end{pmatrix} &= \begin{pmatrix} -p(r(t)) & p(r(t)) & 0 \\ q(r(t)) & -q(r(t)) & \bar{r}(r(t)) \\ 0 & -v(r(t)) & -z(r(t)) \end{pmatrix} \begin{pmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \end{pmatrix} + \begin{pmatrix} -w(r(t))\check{f}(x_{i1}(t), r(t)) \\ 0 \\ 0 \end{pmatrix} \\ &+ \tilde{u}_i(x_i(t) - s(t), r(t)) + \sum_{j=1}^N \bar{C}_{ij}(r(t))M_{ij}(x_i(t) - x_j(t), r(t)) \\ &+ \begin{pmatrix} \bar{v}_{i1}(r(t)) & 0 & 0 \\ 0 & \bar{v}_{i2}(r(t)) & 0 \\ 0 & 0 & \bar{v}_{i3}(r(t)) \end{pmatrix} \dot{B}_i(t), \quad i \in \mathbb{L}, \end{aligned} \quad (20)$$

where  $p(r(t)) = 1/(R(r(t))C_1(r(t)))$ ,  $q(r(t)) = 1/(R(r(t))C_2(r(t)))$ ,  $\bar{r}(r(t)) = 1/C_2(r(t))$ ,  $v(r(t)) = 1/L(r(t))$ ,  $z(r(t)) = R_0(r(t))/L(r(t))$ ,  $w(r(t)) = 1/C_1(r(t))$ . Parameters  $p(r(t))$ ,  $q(r(t))$ ,  $\bar{r}(r(t))$ ,  $v(r(t))$ ,  $z(r(t))$ ,  $w(r(t))$  are positive.  $x_i = (x_{i1}, x_{i2}, x_{i3})^T$ ,  $\bar{v}_{i1}(r(t))$ ,  $\bar{v}_{i2}(r(t))$ ,  $\bar{v}_{i3}(r(t)) \neq 0$ ,  $\check{f}(x_{i1}(t), r(t)) = f(x_{i1}(t))$ , and  $s(t)$  will be explained later.

We let

$$\begin{aligned} \Xi(x_i(t), r(t)) &= \begin{pmatrix} -p(r(t)) & p(r(t)) & 0 \\ q(r(t)) & -q(r(t)) & \bar{r}(r(t)) \\ 0 & -v(r(t)) & -z(r(t)) \end{pmatrix} x_i + \begin{pmatrix} -w(r(t))\check{f}(x_{i1}, r(t)) \\ 0 \\ 0 \end{pmatrix}, \\ \tilde{u}_i(x_i - s, r(t)) &= \begin{pmatrix} o_i^{(k)}(x_{i1} - s_1) \\ \varrho_i^{(k)}(x_{i2} - s_2) \\ \tau_i^{(k)}(x_{i3} - s_3) \end{pmatrix}, \quad \bar{G}_i(r(t)) = \begin{pmatrix} \bar{v}_{i1}(r(t)) & 0 & 0 \\ 0 & \bar{v}_{i2}(r(t)) & 0 \\ 0 & 0 & \bar{v}_{i3}(r(t)) \end{pmatrix}. \end{aligned}$$

Then, system (20) can be modified as:

$$\begin{aligned} dx_i(t) &= \left[ \Xi(x_i(t), r(t)) + \tilde{u}_i(x_i(t) - s(t), r(t)) + \sum_{j=1}^N \bar{C}_{ij}(r(t))M_{ij}(x_i(t) - x_j(t), r(t)) \right] dt \\ &+ \bar{G}_i(r(t))dB_i(t), \quad i \in \mathbb{L}, \end{aligned} \quad (21)$$

where  $s(t) = (s_1(t), s_2(t), s_3(t))^T \in \mathbb{R}^3$  satisfies  $\dot{s}(t) = \Xi(s(t), r(t))$ . System (21) is established on digraph  $\mathcal{H}$  with  $N$  vertices.

Let  $e_i(t) \triangleq x_i(t) - s(t) = (e_{i1}(t), e_{i2}(t), e_{i3}(t))^T$ ,  $\tilde{\Xi}(e_i(t), r(t)) \triangleq \Xi(x_i(t), r(t)) - \Xi(s(t), r(t))$ . We can gain the error system as follows:

$$\begin{aligned} de_i(t) &= \left[ \tilde{\Xi}(e_i(t), r(t)) + \tilde{u}(e_i(t), r(t)) + \sum_{j=1}^N \bar{C}_{ij}(r(t))M_{ij}(e_i(t) - e_j(t), r(t)) \right] dt \\ &+ \bar{G}_i(r(t))dB_i(t), \quad i \in \mathbb{L}. \end{aligned} \quad (22)$$

Then we have the following theorem that guarantees the existence of synchronized stationary distribution for system (20).

**Theorem 4.** *If the following conditions hold, then system (20) admits a synchronized stationary distribution.*

**D1.** *For any  $x \in \mathbb{R}^3$ , there are positive constants  $Q_{ij}(k)$  satisfying*

$$|M_{ij}(x, k)| \leq Q_{ij}(k)|x|, \quad i, j \in \mathbb{L}, \quad k \in \mathbb{S}.$$

**D2.** *For  $p \geq 2$  and each  $i \in \mathbb{L}$ , matrix*

$$\mathcal{B}_i := -\text{diag} \left\{ p\tilde{\alpha}_i^{(1)} + \frac{1}{2}(p-1)(p-2) + D_i^{(1)}, \dots, p\tilde{\alpha}_i^{(l)} + \frac{1}{2}(p-1)(p-2) + D_i^{(l)} \right\} - \mathfrak{T}$$

*is a nonsingular M-matrix, where  $\tilde{\alpha}_i^{(k)} = -\min\{\check{A}(k), \check{B}(k), \check{C}(k)\} + \max\{o_i^{(k)}, \varrho_i^{(k)}, \tau_i^{(k)}\}$ ,  $D_i^{(k)} = \sum_{j=1}^N \frac{1}{2} p\tilde{C}_{ij}(k) + \sum_{j=1}^N 2p\tilde{C}_{ij}(k)Q_{ij}^2(k)$ , in which  $\check{A}(k) = \frac{1}{2R(k)C_1(k)} + \frac{\tilde{m}_{e_{i1}}^{(k)}}{C_1(k)} - \frac{1}{2R(k)C_2(k)}$ ,  $\check{B}(k) = \frac{1}{2R(k)C_2(k)} - \frac{1}{2R(k)C_1(k)} - \frac{1}{2C_2(k)} + \frac{1}{2L(k)}$ ,  $\check{C}(k) = \frac{R_0(k)}{L(k)} - \frac{1}{2C_2(k)} + \frac{1}{2L(k)}$ .*

**D3.** *Digraph  $(\mathcal{H}, (\tilde{C}_{ij}(k))_{N \times N})$  is strongly connected.*

**Proof.** In literature [42], it has been proved that function  $f(\cdot, \cdot)$  satisfies the following equality:  $\check{f}(x_i(t), k) - \check{f}(s(t), k) = \varpi_{e_i}^{(k)}(t)(x_i(t) - s(t))$ , where  $\varpi_{e_i}^{(k)}(t) = (m_{e_i}^{(k)}(t))_{3 \times 3}$  is a bounded matrix, in which the elements  $m_{e_i}^{(k)}(t)$  satisfy  $\underline{m}_{e_i}^{(k)} \leq m_{e_i}^{(k)}(t) \leq \bar{m}_{e_i}^{(k)}$  with  $\underline{m}_{e_i}^{(k)}$  and  $\bar{m}_{e_i}^{(k)}$  being known constants. In particular, for  $x_{i1}(t)$  and  $s_1(t)$ ,  $\check{f}(x_{i1}(t), k) - \check{f}(s_1(t), k) = \tilde{m}_{e_{i1}}^{(k)}(x_{i1}(t) - s_1(t))$  with  $G_{a1} \leq \tilde{m}_{e_{i1}}^{(k)} \leq G_{b1}$ . Thus, we can obtain that the function  $f(x_i(t), k)$  satisfies Lipschitz condition. Moreover, it satisfies linear growth condition obviously. Then there exists a unique solution to system (20). Furthermore, according to  $\check{f}(x_{i1}(t), k) - \check{f}(s_1(t), k) = \tilde{m}_{e_{i1}}^{(k)}(x_{i1}(t) - s_1(t))$ , we have

$$\begin{aligned} e_i^T \tilde{\Xi}(e_i, k) &= e_i^T (\tilde{\Xi}(x_i(t), k) - \tilde{\Xi}(s(t), k)) \\ &\leq \left( -\frac{1}{R(k)C_1(k)} - \frac{\tilde{m}_{e_{i1}}^{(k)}}{C_1(k)} \right) |e_{i1}|^2 - \frac{1}{R(k)C_2(k)} |e_{i2}|^2 - \frac{R_0(k)}{L(k)} |e_{i3}|^2 \\ &\quad + \left( \frac{1}{2R(k)C_1(k)} + \frac{1}{2R(k)C_2(k)} \right) (|e_{i1}|^2 + |e_{i2}|^2) \\ &\quad + \left( \frac{1}{2C_2(k)} - \frac{1}{2L(k)} \right) (|e_{i2}|^2 + |e_{i3}|^2) \\ &= - \left( \frac{1}{2R(k)C_1(k)} + \frac{\tilde{m}_{e_{i1}}^{(k)}}{C_1(k)} - \frac{1}{2R(k)C_2(k)} \right) |e_{i1}|^2 \\ &\quad - \left( \frac{1}{2R(k)C_2(k)} - \frac{1}{2R(k)C_1(k)} - \frac{1}{2C_2(k)} + \frac{1}{2L(k)} \right) |e_{i2}|^2 \\ &\quad - \left( \frac{R_0(k)}{L(k)} - \frac{1}{2C_2(k)} + \frac{1}{2L(k)} \right) |e_{i3}|^2 \\ &\leq - \min \{ \check{A}(k), \check{B}(k), \check{C}(k) \} |e_i|^2, \end{aligned} \tag{23}$$

where  $\check{A}(k)$ ,  $\check{B}(k)$  and  $\check{C}(k)$  are exhibited in condition D2. Meanwhile,

$$e_i^T \tilde{u}_i(e_i, k) = o_i^{(k)} |e_{i1}|^2 + \varrho_i^{(k)} |e_{i2}|^2 + \tau_i^{(k)} |e_{i3}|^2 \leq \max \{ o_i^{(k)}, \varrho_i^{(k)}, \tau_i^{(k)} \} |e_i|^2.$$

Therefore, in view of condition  $D1$ , condition (i) in [Theorem 2](#) is satisfied. Conditions (ii) and (iii) are checked by means of conditions  $D2$  and  $D3$ , respectively. We can see that all the conditions in [Theorem 2](#) hold. Therefore, system (20) exists a synchronized stationary distribution according to [Theorem 2](#).  $\square$

**Remark 6.** Chua's circuit has complicated nonsmooth dynamical behaviors that receive more research attention. Many synchronization problems are studied in the Chua's circuits network. For example, Li et al. investigated fuzzy approximation-based global pinning synchronization control of Chua's circuits network in [\[32\]](#). And chaos synchronization in Chua's circuit was studied in [\[41\]](#). Compared with the above literature, we study synchronized stationary distribution of the Chua's circuits network, which is a kind of synchronization weaker than the usual form. Up till now, there are few results about this aspect of Chua's circuit.

#### 4.4. A numerical simulation for Chua's circuits network

Then, we give a numerical example to illustrate the theoretical results of the application to the network of Chua's circuits. Let  $N = 4$ , we consider the existence of synchronized stationary distribution for system (20) according to system (22). Let state space  $\mathbb{S} = \{1, 2\}$  with generator

$$\mathcal{T} = (\pi_{kh})_{2 \times 2} = \begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix}.$$

Meanwhile, the functions are chosen as follows:

$$M_{ij}(x_i - x_j, 1) = 0.2(x_i - x_j), \quad M_{ij}(x_i - x_j, 2) = 0.4(x_i - x_j), \quad i, j = 1, 2, 3, 4.$$

And set  $Q_{ij}(1) = 0.25$ ,  $Q_{ij}(2) = 0.45$ ,  $i, j = 1, 2, 3, 4$ . Therefore, condition  $D1$  holds. Then, we choose

$$(\bar{C}_{ij}(1))_{4 \times 4} = \begin{pmatrix} 0 & 0.2 & 0.1 & 0 \\ 0.3 & 0 & 0 & 0.2 \\ 0.1 & 0.3 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \end{pmatrix}, \quad (\bar{C}_{ij}(2))_{4 \times 4} = \begin{pmatrix} 0 & 0.2 & 0.1 & 0 \\ 0.1 & 0 & 0 & 0.1 \\ 0 & 0.1 & 0 & 0 \\ 0.4 & 0.2 & 0 & 0 \end{pmatrix}.$$

Obviously, digraphs  $(\mathcal{H}, (\bar{C}_{ij}(k))_{4 \times 4})$ ,  $k = 1, 2$  are strongly connected. We set  $p = 2$ ,  $R(k) = 1$ ,  $C_1(k) = 0.5$ ,  $C_2(k) = 0.9$ ,  $L(k) = 0.1$ ,  $R_0(k) = 0.2$ ,  $G_{a1} = -0.7559$ ,  $G_{b1} = -0.10$ ,  $\tilde{m}_{e_{11}}^{(k)} = -0.1572$ ,  $i = 1, 2, 3, 4$ ,  $k = 1, 2$ . So we can calculate that  $\check{A}(k) = 0.130$ ,  $\check{B}(k) = 4$ ,  $\check{C}(k) = 6.444$ . Let  $\bar{v}_{ij}(k) = 2.5$ ,  $i = 1, 2, 3, 4$ ,  $j = 1, 2, 3$ ,  $k = 1, 2$ . Other parameters can be selected as follows for  $k = 1, 2$ :

$$\begin{aligned} o_1^{(k)} &= -0.15, & o_2^{(k)} &= -0.25, & o_3^{(k)} &= -0.35, & o_4^{(k)} &= -0.45, \\ \varrho_1^{(k)} &= -0.15, & \varrho_2^{(k)} &= -0.25, & \varrho_3^{(k)} &= -0.35, & \varrho_4^{(k)} &= -0.45, \\ \tau_1^{(k)} &= -0.15, & \tau_2^{(k)} &= -0.25, & \tau_3^{(k)} &= -0.35, & \tau_4^{(k)} &= -0.45. \end{aligned}$$

According to the parameters selected above, we can get the following values:

$$\begin{aligned}\tilde{\alpha}_1^{(k)} &= -0.28, & \tilde{\alpha}_2^{(k)} &= -0.38, & \tilde{\alpha}_3^{(k)} &= -0.48, & \tilde{\alpha}_4^{(k)} &= -0.58, & k &= 1, 2, \\ D_1^{(1)} &= 0.425, & D_1^{(2)} &= 0.543, & D_2^{(1)} &= 0.625, & D_2^{(2)} &= 0.362, \\ D_3^{(1)} &= 0.500, & D_3^{(2)} &= 0.181, & D_4^{(1)} &= 0.625, & D_4^{(2)} &= 1.086.\end{aligned}$$

Therefore, we can obtain that

$$\mathcal{B}_1 = -\text{diag}\{-0.135, -0.017\} - \mathfrak{T} = \begin{pmatrix} 3.135 & -3 \\ -1 & 1.017 \end{pmatrix},$$

$$\mathcal{B}_2 = -\text{diag}\{-0.135, -0.398\} - \mathfrak{T} = \begin{pmatrix} 3.135 & -3 \\ -1 & 1.398 \end{pmatrix},$$

$$\mathcal{B}_3 = -\text{diag}\{-0.46, -0.779\} - \mathfrak{T} = \begin{pmatrix} 3.46 & -3 \\ -1 & 1.779 \end{pmatrix},$$

$$\mathcal{B}_4 = -\text{diag}\{-0.535, -0.074\} - \mathfrak{T} = \begin{pmatrix} 3.535 & -3 \\ -1 & 1.074 \end{pmatrix},$$

which are nonsingular M-matrices distinctly. Thus, condition D2 in Theorem 4 is also satisfied. In summary, all conditions in Theorem 4 are verified. Then, system (20) admits a synchronized stationary distribution. Furthermore, the initial values are selected as follows:

$$e_0 = (10.9, -11.5, 4.6, -13.1, 8.5, -15.4, 8.9, -6.5, 5.6, -11.1, 8.5, 10.6)^T.$$

We can get the figure of the distribution of sample path in the phase space shown in Fig. 4. This numerical result shows the effectiveness of our theoretical results. Moreover, if we change perturbation intensity  $\bar{v}_{ij}(k) = 10.5, i = 1, 2, 3, 4, j = 1, 2, 3, k = 1, 2$ , the distribution of sample path is shown in Fig. 5, which has the larger stationary distribution region than the counterpart in Fig. 4. On the other hand, we change the integral parts of the control coefficients into 10, and the corresponding stationary distribution region (see Fig. 6) is obviously smaller than that in Fig. 4. Besides, from Fig. 7, we can see that the distribution of sample path for system (22) eventually goes to zero, which implies that system (20) achieves complete synchronization. These comparisons explain Remark 3 intuitively.

## 5. Conclusion

The problem of synchronized stationary distribution for HSCSs was considered. Based on Lyapunov method and graph theory, two novel synchronized stationary distribution criteria have been established for the considered HSCSs. It has been shown that this kind of synchronization is weaker than the usual form in existing results. And the effectiveness has been shown through applications to oscillators and Chua's circuits with numerical simulations being carried out. Since discontinuous control has more advantages than continuous control, in the future, we will consider discontinuous control instead of feedback control, including intermittent control, impulsive control and so on, to achieve synchronized stationary distribution. At the same time, the directed graphs in practical applications do not always satisfy the conditions of strong connectivity, so future work may be done trying to remove the strong connectedness of the digraph. And we find that fuzzy coupled systems show its potential advantages in image processing and pattern recognition, which have attracted considerable attention of many scholars [43–46]. Therefore, next, we will consider the coupling system based on fuzzy graph to study the synchronized stationary distribution of fuzzy coupled systems.



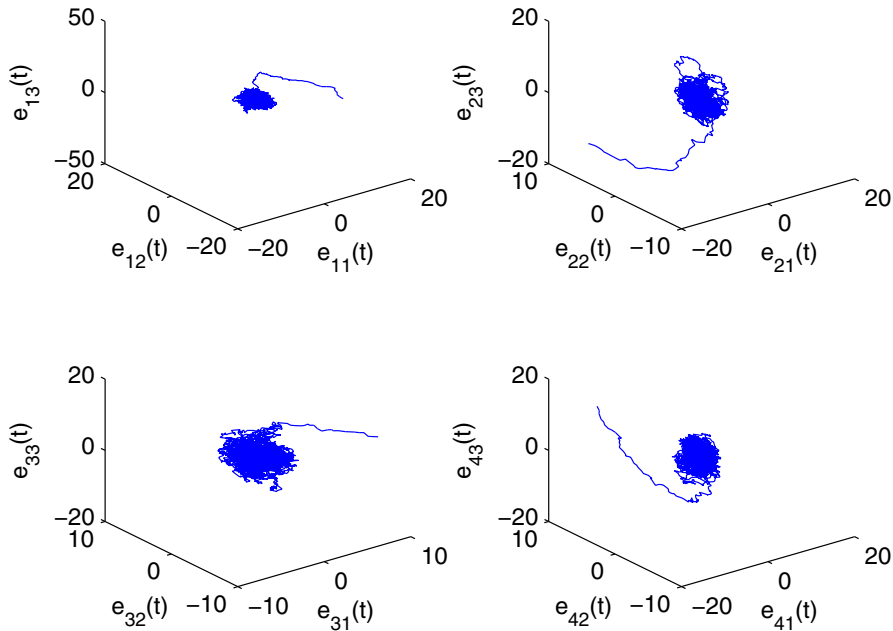


Fig. 4. The distribution of sample path of system (22).

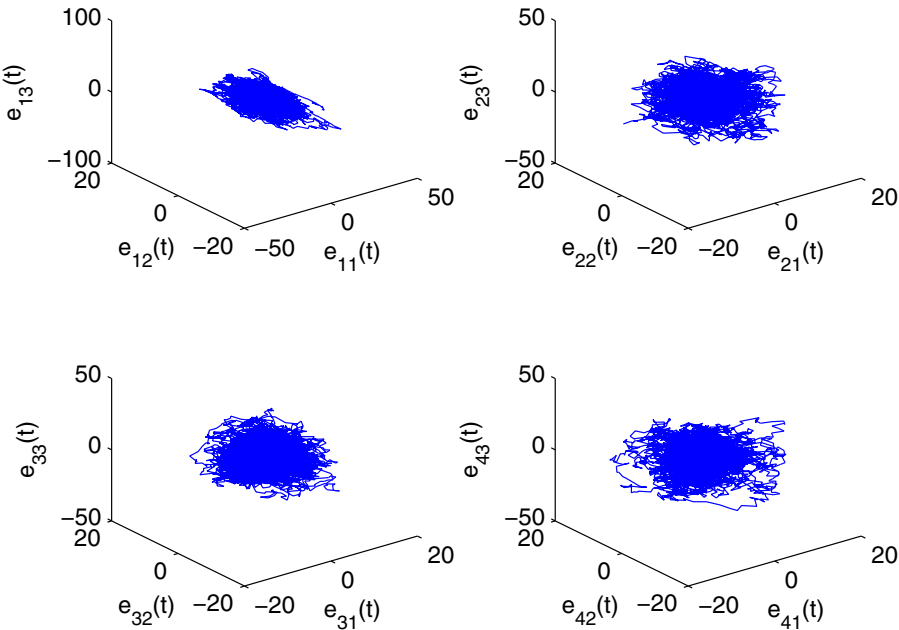


Fig. 5. The distribution of sample path of system (22) under the larger perturbation intensity.

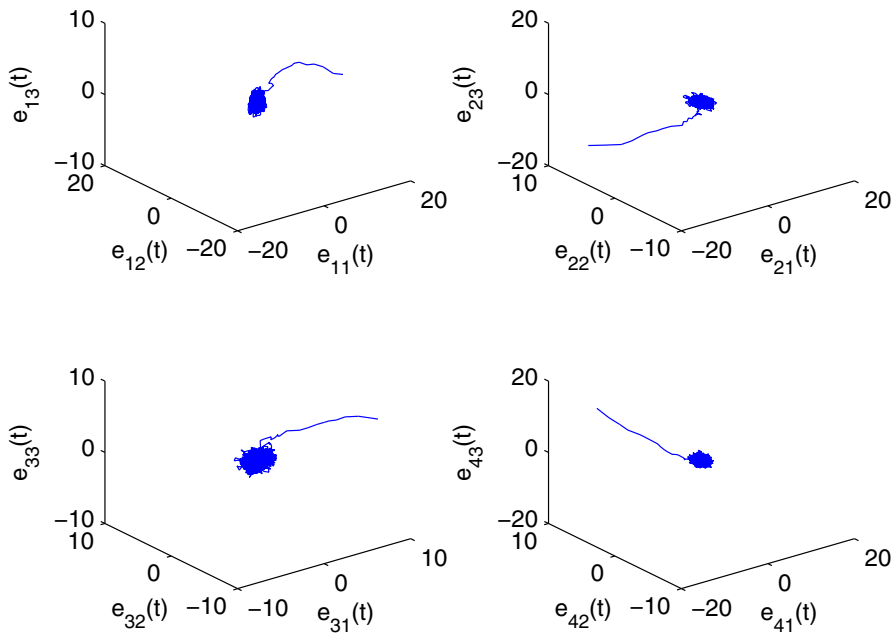


Fig. 6. The distribution of sample path of system (22) under the larger control coefficients.

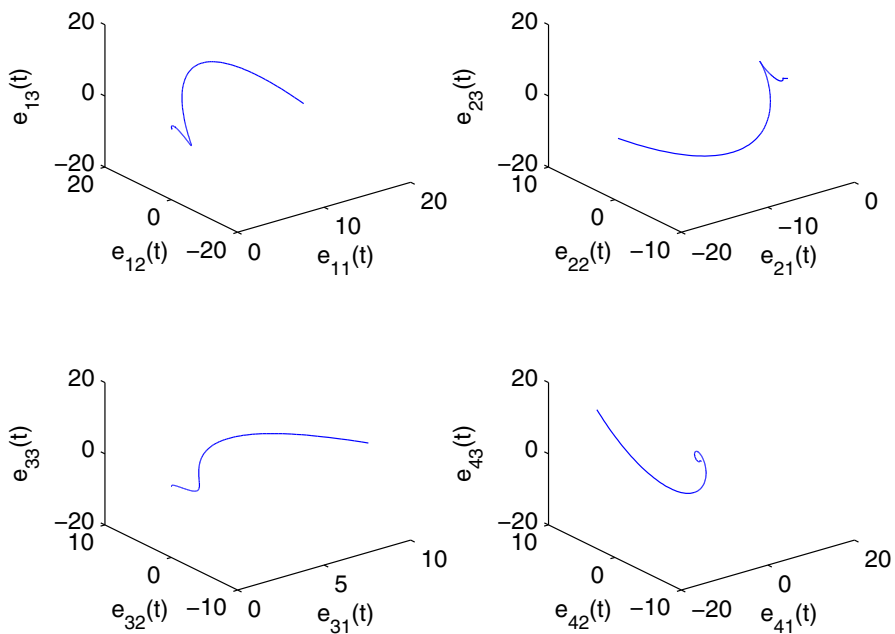


Fig. 7. The distribution of sample path of system (22) without perturbation.

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